The Idea

R. Rebonato

EDHEC Business School
corresponding author: riccardo.rebonato@edhec.edu,
10 Fleet Place, London EC4M 7RB

April 16, 2026

Abstract

We

1 The Set-Up

For simplicity we consider zero-coupon bonds (real and nominal). We denote with upper case quantities for nominal bonds (eg, P_0^T or Y_0^T for the price and yield, respectively) and use same-letter symbols with lower case for real quantities (eg, p_0^T or y_0^T). Instantaneous inflation is denoted by $\pi(t)$.

We consider three instruments:

1. a zero-coupon index-linked real bond (the TIPS);
2. an inflation swap (the IS);
3. a nominal bond (the Nominal).

A unit notional of TIPS pays at maturity, T

$$\exp\left[\int_0^T \pi(s)ds\right] \tag{1}$$

A unit notional of IS pays at maturity, T,

$$\exp\left[f_0^T \cdot (T - t_0)\right] - \exp\left[\int_0^T \pi(s)ds\right] \tag{2}$$

A portfolio, Π_0^T, made up one one unit of TIPS and one unit of IS pays at maturity

$$\Pi_T^T = \underbrace{\exp\left[\int_0^T \pi(s)ds\right]}_{TIPS} + \underbrace{\exp\left[f \cdot (T - t_0)\right] - \exp\left[\int_0^T \pi(s)ds\right]}_{IS} = \exp\left[f_0^T \cdot (T - t_0)\right] \tag{3}$$

The payment at time T is certain and the same in all states of the world.

A T-maturity nominal bond pays at time T \$1 with certainty and in all states of the world. Therefore $\exp\left[-f_0^T \cdot T\right]$ units of the TIPS + IS combinations also pay \$1 with certainty and in all states of the world. *Assuming that all positions can be held with no constraints to maturity*, by the law of one price $\exp\left[-f_0^T \cdot T\right]$ units of the TIPS + IS combination must cost today the same as the notional bond, P_0^T. Since entering the IS entails no cost, it must be that

$$\exp\left[-f_0^T \cdot T\right] \cdot p_0^T = P_0^T \rightarrow p_0^T = P_0^T \cdot \exp\left[-f_0^T \cdot T\right] \tag{4}$$

Writing

$$P_0^T \equiv \exp\left[-Y_0^T \cdot T\right] = \exp\left[-(y_0^T + f_0^T) \cdot T\right] \tag{5}$$

we have that, to satisfy the law of one price, the price of the TIPS must be

$$p_0^T = \exp\left[-y_0^T \cdot T\right] \tag{6}$$

It is simplest, however, to consider the portfolio, Π_0^T, as a security with a payoff identical to that of the nominal ZCB. If the law of one price is respected, the security also has a yield $Y(\Pi_0^T) = Y_0^T$.

Suppose that, for whatever reason, the market price today of the portfolio, Π_0^T, is less than the price of the nominal ZCB:

$$\Pi_0^T < P_0^T \tag{7}$$

This lower price is reflected in a higher yield,

$$Y(\Pi_0^T) = Y_0^T + \lambda_0^T \tag{8}$$

with $\lambda_0^T > 0$.

If the portfolio, Π_0^T, were a complex derivative security, it would be customary to upfront the value of the long position in the portfolio minus the nominal ZCB, giving an immediate profit today, $PV(0)$ of

$$PV(0) = \exp\left[-Y_0^T \cdot T\right] \cdot \left(1 - \exp\left[-\lambda_0^T \cdot T\right]\right) > 0 \tag{9}$$

However, all the components of the strategy are securities with transparent market prices, transacted at market rates. So, in a mark-to-market environment the whole strategy (including the net cash from the sale of the expensive nominal ZCB and the purchase of the cheap portfolio) has zero value today. The net cash received today, $cash(0)$, is, by definition,

$$cash(0) = \exp\left[-Y_0^T \cdot T\right] \cdot \left(1 - \exp\left[-\lambda_0^T \cdot T\right]\right) \tag{10}$$

This cash can be invested at th e rate Y_0^T from time 0 to time T, to yield at time T

$$cash(T) = \left(1 - \exp\left[-\lambda_0^T \cdot T\right]\right) \tag{11}$$

At any time t, $0 \leq t \leq T$ the portfolio of securities (without the cash) is worth

$$\Pi_t^T = \exp\left[-Y_t^T \cdot \tau\right] \left(\exp\left[-\lambda_t^T \cdot \tau\right] - 1\right) \tag{12}$$

The value of the loan at time t is equal to

$$cash(t) = \left(\exp\left[-Y_0^T \cdot T\right] \cdot \left(1 - \exp\left[-\lambda_0^T \cdot T\right]\right)\right) \cdot \exp[Y_0^T \cdot t] \tag{13}$$

or

$$cash(t) = \exp\left[-Y_0^T \cdot \tau\right] \cdot \left(1 - \exp\left[-\lambda_0^T \cdot T\right]\right) \tag{14}$$

with $\tau \equiv T - t$. So, the time-t value of the whole strategy, $S(t)$, is given by

$$\begin{aligned} S(t) = \\ + \exp\left[-Y_0^T \cdot \tau\right] - \exp\left[-Y_t^T \cdot \tau\right] \\ + \exp\left[-Y_t^T \cdot \tau\right] \exp\left[-\lambda_t^T \cdot \tau\right] - \exp\left[-Y_0^T \cdot \tau\right] \exp\left[-\lambda_0^T \cdot \tau\right] \end{aligned} \tag{15}$$

or

$$\begin{aligned} S(t) = \\ + \exp\left[-Y_0^T \cdot \tau\right] (1 - \exp\left[-\lambda_0^T \cdot \tau\right]) \\ - \exp\left[-Y_t^T \cdot \tau\right] (1 - \exp\left[-\lambda_t^T \cdot \tau\right]) \end{aligned} \tag{16}$$

Note that the first line $(+\exp\left[-Y_0^T \cdot \tau\right] (1 - \exp\left[-\lambda_0^T \cdot \tau\right]))$ is a deterministic function of time. The second line $(-\exp\left[-Y_t^T \cdot \tau\right] (1 - \exp\left[-\lambda_t^T \cdot \tau\right]))$, however, is stochastic.The whole strategy will have a more positive (negative) value at time t if nominal yields, Y_t^T, have increased (decreased) and the spread, λ_t^T, has decreased (increased).

In the case of an instantaneous change in λ_0^T at time 0 to a value $\hat{\lambda}_0^T$, we have for the value of the whole strategy

$$\hat{S}(0) = \exp\left[-Y_0^T \cdot T\right] (\exp\left[-\hat{\lambda}_0^T \cdot \tau\right] - \exp\left[-\lambda_0^T \cdot \tau\right]) \tag{17}$$

2 Limits to Arbitrage

We want to quantify the effects of limits to arbitrage.

Since all the stochastic variables in the value equation above can change, before expiry the strategy X_t can have a negative value. If the negative value exceeds a certain threshold, the trader is forced to unwind the strategy at a loss. The full distribution of values for the strategy is therefore made up of

1. a Dirac-delta centred at $P_0^T \cdot 1 - \exp\left[\lambda_0^T \cdot T\right] > 0$ and with probability mass equal to the fraction of the paths that reached T without a forced unwinding; plus

2. a continuous distribution of negative X_t, with each value equal to the discounted value of the strategy at the forced unwinding time.

The overall distribution will in general have a positive expectation, but the risk averse investor will not pay this expected value. Instead, she will pay the certainty equivalent of her initial wealth plus the overall distribution (a certainty equivalent that depends on her risk aversion). If the CE is higher than her initial wealth, she will enter the strategy; otherwise she will not.

What is her 'initial wealth'? I would argue that it should be set equal to the capital charge (evaluated at, say, multiplier times VaR) associated with the strategy (it should be the marginal VaR, really, but this becomes difficult). We can discuss here.

References